

# **Requirement Specification**

## **1. The Purpose of the Project**

a) The BoatLoop project aims to develop a boat-sharing system that allows private boat owners to rent their boats to passengers seeking travel or recreational experiences. This system aims to provide a seamless, secure, and efficient platform for boat rentals, enhancing accessibility to water travel and ensuring compliance with maritime regulations.

b) The Goal of the Project:

- Goal: To launch the BoatLoop system by May 2025 within a budget of 15 million CAD.
- Measurement: Success will be measured by the system's ability to handle up to 10,000 concurrent users, maintain an uptime of 98%, and ensure data security and compliance with maritime regulations.
- Advantage: The system will provide boat owners with additional income opportunities and passengers with a wide range of boating experiences. It will also ensure a secure and transparent experience, building user trust.

## **2. The Stakeholders**

a) The Client: TrueNorth Voyages, the company developing the BoatLoop system.

b) The Customer: Passengers seeking boat rentals for travel or recreational purposes.

c) Other Stakeholders:

- Boat Owners: Individuals or businesses listing their boats for rent.
- Maritime Authorities: Regulators ensuring compliance with maritime laws and passenger safety.
- Customer Support Agents: Assisting users with bookings, cancellations, and other services.
- Development Team: Responsible for designing, developing, and maintaining the platform.
- Marketing Experts: Promoting the platform to potential users.
- Legal Experts: Ensuring the platform complies with relevant laws and regulations.
- Usability Experts: Ensuring the platform is user-friendly and accessible.
- Testers: Ensuring the platform functions correctly and meets requirements.
- System Engineers and Software Engineers: Developing and maintaining the technical infrastructure.
- Business Analysts: Analyzing business requirements and ensuring they are met.

d) The Hands-on Users of the Product:

- Passengers: Using the platform to book boat trips.
- Boat Owners: Using the platform to list and manage their boats.

e) Priorities Assigned to Users:

- Key Users: Passengers and boat owners, as their satisfaction is critical to the platform's success.
- Secondary Users: Customer support agents and marketing experts, as their roles support the primary users.
- Unimportant Users: Infrequent or unauthorized users, whose opinions have little impact on the platform's long-term success.

f) Maintenance Users and Service Technicians:

- Maintenance Users: System administrators responsible for maintaining the platform.
- Service Technicians: Technical support staff addressing any technical issues.

### **3. Project Constraints**

a) Solution Constraints:

- The platform must be accessible on both desktop and mobile devices.
- The system must use encryption for all data transmissions and store passwords securely.

b) Partner or Collaborative Applications:

- Integration with payment gateways for secure transactions.
- Compliance with maritime regulations databases.

c) Off-the-Shelf Software:

- Use of existing software solutions for payment processing and data encryption.

d) Schedule Constraints:

- The system must be launched by May 2025.

e) Budget Constraints:

- The project has a set budget of 15 million CAD.

### **4. Naming Conventions and Terminology**

- Passenger: An individual using the platform to book a boat for leisure or travel on the water.
- Boat-Sharing System: A digital service linking private boat owners with renters seeking short-term boating experiences.
- TrueNorth Voyages: The innovative company offering a boat-sharing app and service.
- Boat Owner: A registered user who lists their boat for hire by passengers through the platform.
- Booking System: The digital feature that enables passengers to find, schedule, and confirm boat rentals seamlessly.
- Trip: A planned water-based excursion rented through the platform, chosen by the passenger.

- Boat Specification: Key information about a vessel's size, features, and amenities provided by the owner.
- Payment Processing: The secure technology that handles and confirms payment transactions for boat rentals.
- Cancellation Fees: Charges are applied to passengers if they plan to cancel a booked trip.
- Listing: A published profile of a boat available for rent, including details, photos, and pricing.
- Testimonial: Personal written reflections from passengers on their boating experience.
- Ratings: A feedback score passengers give after a trip, reflecting their overall satisfaction.
- Agents: On-ground customer service specialists who help users with booking queries and trip logistics.
- Maritime Rules and Regulations: Compliance standards the platform must follow to ensure safety and legal adherence in maritime operations.
- Data Protection: Strategies and systems in place to ensure users' personal and payment information remains secure.
- System Performance: The platform's ability to deliver fast, reliable services like quick searches, bookings, and transactions.

## 5. Relevant Facts and Assumptions

### a) Relevant Facts:

- The system must handle up to 10,000 concurrent users.
- The system must maintain an uptime of 98%.

### b) Business Rules:

- The system must comply with GDPR and other relevant data protection regulations.
- The system must ensure compliance with maritime regulations for all bookings.

### c) Assumptions:

- Necessary legal permissions will be obtained.
- Adequate insurance coverage will be secured.
- The system will be developed within the specified budget and timeframe.

## Functional Requirements:

- Creating the Account:
  - Description: The system shall allow boat owners and passengers to create accounts by providing personal details, verification documents, and a password.
  - Rationale: To enable users to access and use the platform.
  - Acceptance Criteria: Account creation is successful, and verification is completed.

- Priority: High

- Boat Listing:

- Description: The system shall allow boat owners to list their boats with specifications, trip details, photos, and services offered.
- Rationale: To provide detailed information about available boats to passengers.
- Acceptance Criteria: Boat listing is published and visible to passengers.
- Priority: High

- Selecting a Boat:

- Description: The system shall allow passengers to search for and select boats based on their preferences and trip details.
- Rationale: To enable passengers to find and book suitable boats.
- Acceptance Criteria: Available boats are displayed based on search criteria.
- Priority: High

- Payment:

- Description: The system shall process payments securely using debit/credit card details.
- Rationale: To confirm bookings and handle transactions.
- Acceptance Criteria: Payment and booking confirmations are sent to users.
- Priority: High

- Trip Completion:

- Description: The system shall handle trip completion, collect feedback, and update profiles.
- Rationale: To finalize trips and gather user feedback.
- Acceptance Criteria: Payments are sent to boat owners, and reviews are displayed.
- Priority: Medium

- Customer Support:

- Description: The system shall provide customer support for bookings, cancellations, and other services.
- Rationale: To assist users and resolve issues.

- Acceptance Criteria: Issues are resolved, and user needs are met.
- Priority: Medium

- Data Security:

- Description: The system shall ensure data security with password protection and automatic logout after inactivity.
- Rationale: To protect user data and privacy.
- Acceptance Criteria: Inactivity status is displayed, and data is secure.
- Priority: High

- Boat Owner's Weekly Report:

- Description: The system shall generate weekly reports for boat owners with revenue, booking data, and trip details.
- Rationale: To provide boat owners with performance insights.
- Acceptance Criteria: Detailed reports are generated and accessible.
- Priority: Medium

- Maritime and Port Authority Regulations:

- Description: The system shall comply with maritime regulations and display permission statuses.
- Rationale: To ensure legal compliance and safety.
- Acceptance Criteria: Permissions and notices are displayed as required.
- Priority: High

- Cancellation (Passenger):

- Description: The system shall handle trip cancellations by passengers and process refunds.
- Rationale: To manage cancellations and refunds.
- Acceptance Criteria: Cancellation window and refund status are displayed.
- Priority: Medium

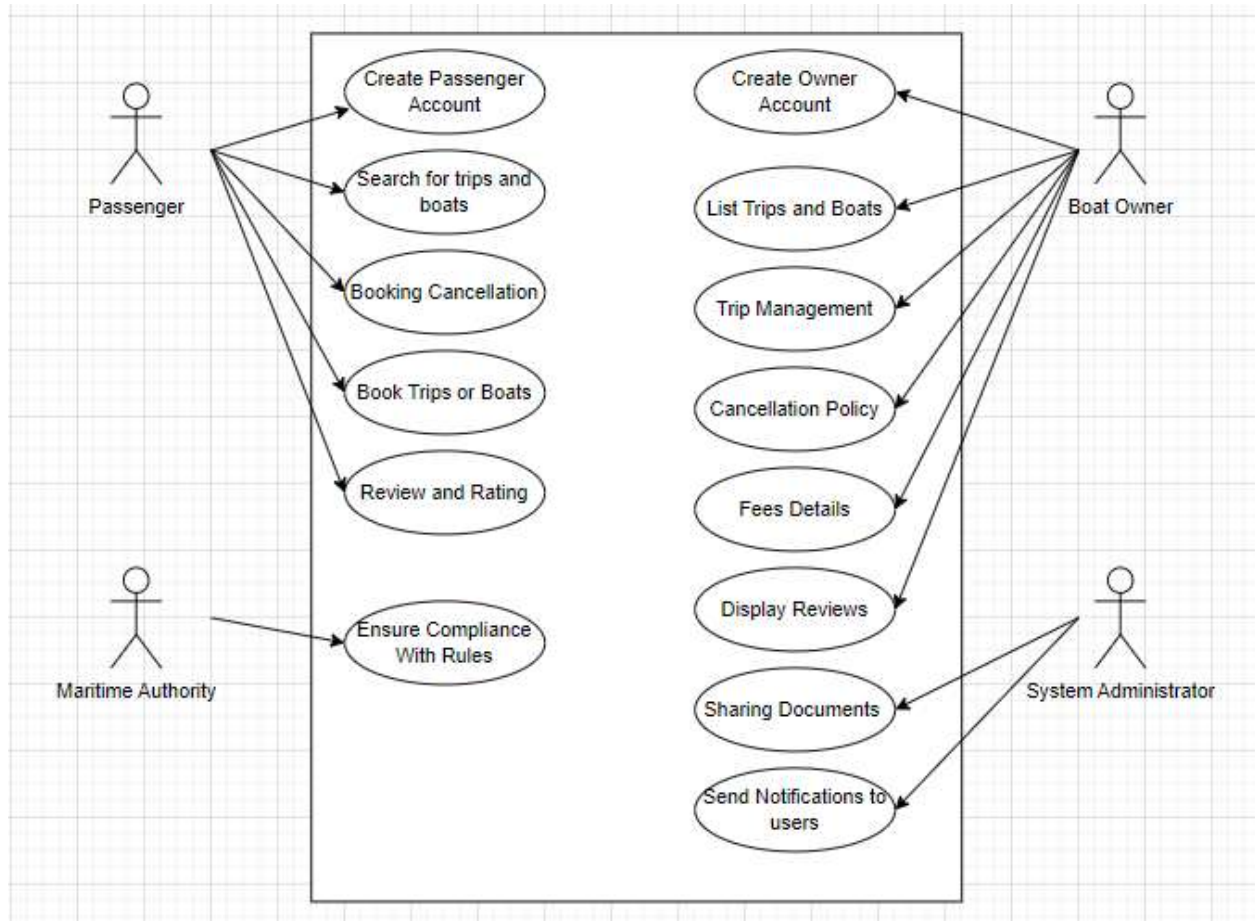
- Cancellation (Boat Owner):

- Description: The system shall handle cancellations by boat owners and apply relevant fees.
- Rationale: To manage cancellations and enforce policies.
- Acceptance Criteria: Policy and fees are displayed.
- Priority: Medium

#### **Non-Functional Requirements:**

- Performance Requirements: The system shall handle up to 10,000 concurrent users with a response time of less than 2 seconds.
- Security Requirements: The system shall use encryption for all data transmissions and store passwords securely.
- Usability Requirements: The system shall have an intuitive user interface accessible on both desktop and mobile devices.
- Reliability Requirements: The system shall have an uptime of 98% and provide regular backups.
- Compliance Requirements: The system shall comply with GDPR and other relevant data protection regulations.

## **Product Use Case Diagram**



## Product Use Case Scenarios

### 1. Management of User Accounts

**Trigger:** A user starts the process of creating an account.

**Preconditions:** The user enters accurate contact details and personal information.

**Interested Stakeholders:** Customer service representatives, boat owners, and passengers

**Actor:** User

1. The user submits personal details (name, email, phone number).
2. The system verifies the information
3. Then It sends a confirmation email.
4. The user sets up a password.
5. Then enter the same passwords to verify the previous password is the same.
6. The system confirms account creation.

7. The user logs in for the first time.

**Outcome:** The user account was successfully created and verified.

## 2. Boat Listing

**Trigger:** The boat owner adds a new boat for listing.

**Preconditions:** The owner has a verified account and provides the necessary boat details.

**Interested Stakeholders:** Boat owners, passengers, and customer support agents.

**Actor:** Boat Owner

1. The owner logs into the system.
2. The system validates the user.
3. The owner will select to "Add a Boat."
4. Owner enters specifications (size, amenities, photos).
5. The system validates the inputs.
6. The system publishes the listing.
7. Listing becomes visible to passengers.

**Outcome:** The boat listing is live and available for booking.

## 3. Trip Search

**Trigger:** A passenger searches for available boats.

**Preconditions:** Passenger provides travel dates and preferred locations.

**Interested Stakeholders:** Passengers, and boat owners.

**Actor:** Passenger

1. A passenger enters trip details (date, destination).
2. The system filters and displays available boats.
3. Passenger reviews boat details.
4. The passenger selects a boat.
5. The system checks for availability.
6. The system confirms availability.

**Outcome:** The passenger is presented with available boat options for booking.



## 4. Booking Confirmation

**Trigger:** Passenger confirms a booking.

**Preconditions:** The passenger selects an available boat and provides payment details.

**Interested Stakeholders:** Passengers, and boat owners.

**Actor:** Passenger

1. The passenger selects "Book Now."
2. The system validates all the information.
3. The system processes payment details.
4. The system generates booking confirmation.
5. Notifications were sent to both passengers and boat owners.
6. The booking status is updated in the system.

**Outcome:** Booking successfully confirmed.

## 5. Payment Processing

**Trigger:** Payment is initiated for a booking.

**Preconditions:** Passenger enters valid payment details.

**Interested Stakeholders:** Passengers, boat owners, financial institutions.

**Actor:** Passenger

1. The passenger enters payment details.
2. The system verifies payment information.
3. The system asks for confirmation before the payment is credited.
4. Then the system processes the transaction securely.
5. Confirmation is sent to the passenger.
6. Payment status is updated in the system.

**Outcome:** Payment successfully processed and recorded.

## 6. Cancellation by Passenger

**Trigger:** Passenger cancels a trip.

**Preconditions:** Cancellation policy allows cancellation within a specified time frame.

**Interested Stakeholders:** Passengers, boat owners, and customer support agents.

**Actor:** Passenger

1. The passenger selects "Cancel Booking."
2. The system checks whether any cancellation fee is applicable.
3. Then the system calculates applicable cancellation fees.
4. The system processes refunds (if applicable).
5. Notification sent to the boat owner.
6. The booking status was updated as cancelled.

**Outcome:** The trip is cancelled with applicable refunds or penalties.

## 7. Cancellation by Boat Owner

**Trigger:** The boat owner cancels a booking.

**Preconditions:** Cancellation aligns with the policy provided by the platform.

**Interested Stakeholders:** Boat owners, passengers, and customer support agents.

**Actor:** Boat Owner

1. The owner selects "Cancel Booking."
2. The system checks for penalties.
3. Then the system calculates penalties (if applicable).
4. A refund is initiated to the passenger.
5. Notifications were sent to both parties.
6. The booking status was updated as cancelled.

**Outcome:** The trip is cancelled with appropriate refunds or penalties.

## 8. Feedback and Reviews

**Trigger:** Passenger or boat owner submits feedback after a trip.

**Preconditions:** The trip must be marked as completed.

**Interested Stakeholders:** Passengers, boat owners, and platform administrators.

**Actor:** Passenger/Boat Owner

1. The user navigates to the "Feedback" section.
2. The user rates the experience and provides comments.
3. The system records and publishes feedback.
4. Feedback updates the boat or passenger profile.
5. The system generates summary reports on reviews.

**Outcome:** Feedback is recorded, improving platform credibility and experience.

## 9. Data Security and Privacy

**Trigger:** User logs into the system.

**Preconditions:** The user has valid credentials.

**Interested Stakeholders:** Users, and platform administrators.

**Actor:** User

1. The user enters a username and password.
2. The system verifies credentials.
3. The system checks if Multi-factor authentication can be triggered.
4. If yes Multi-factor authentication is triggered.
5. The secure session is initiated.
6. Session logs are created for auditing.

**Outcome:** The user accesses the system securely with encrypted data protection.

## 10. Customer Support Interaction

**Trigger:** The user raises a support ticket.

**Preconditions:** The user has an active account.

**Interested Stakeholders:** Passengers, boat owners, and customer support agents.

**Actor:** User

1. The user selects "Contact Support."
2. The user describes the issue and submits a ticket.
3. The support agent reviews.
4. Then, the support agent gives the response.
5. The user and agent resolve the issue collaboratively.

6. The ticket is marked as resolved.

**Outcome:** The user's issue is addressed effectively.

## 11. Regulatory Compliance

**Trigger:** Maritime authority requests passenger data for compliance checks.

**Preconditions:** Request aligns with local regulations.

**Interested Stakeholders:** Maritime authorities, and platform administrators.

**Actor:** Maritime Authority

1. The authority submits a compliance request.
2. The system retrieves relevant data securely.
3. Data is encrypted.
4. Then the data is shared with authorities.
5. Confirmation was sent to both parties.
6. Records updated to reflect compliance.

**Outcome:** Compliance is verified and logged.

## 12. Weekly Revenue Reports for Boat Owners

**Trigger:** Weekly revenue summary generated for boat owners.

**Preconditions:** At least one trip completed during the week.

**Interested Stakeholders:** Boat owners, and platform administrators.

**Actor:** Boat Owner

1. The system compiles trips.
2. The system shows the revenue data for the week.
3. A summary is formatted into a report.
4. The report is sent to the boat owner via email.
5. Data is archived for future reference.
6. The owner reviews the report for insights.

**Outcome:** The boat owner receives a detailed revenue summary for financial tracking.

# User Stories

## Non-Functional Requirements User Stories

### 1. Look and Feel Requirements

- **User Story:** *As a user*, I want the product interface to align with the company's branding standards so that it maintains a consistent visual identity across platforms and builds trust with both customers and users.
  - *Reference:* This requirement stems from the need for a cohesive brand experience, ensuring user familiarity and confidence in the product.

### 2. Usability and Humanity Requirements

- **User Story:** *As a new user*, I want the product to have a straightforward onboarding process with step-by-step guidance so that I can quickly learn how to use it without feeling overwhelmed (RFP Point 2).
  - *Reference:* In previous assignments, we discussed ease of use as a primary factor for user adoption, which is reinforced here with an onboarding focus.
- **User Story:** *As a user*, I want the product to offer interactive prompts and contextual help within the interface so that I can easily navigate through tasks and solve problems without searching through documentation.
  - *Reference:* This ties back to the importance of user-centered design discussed in Assignment 2, where minimizing cognitive load and ensuring users can perform tasks intuitively was a key concern.

### 3. Performance Requirements

- **User Story:** *As a user*, I want the system to maintain fast performance during peak traffic periods so that I am not interrupted or delayed by slow response times (RFP Point 8 - Fault tolerance).
  - *Reference:* This connects to system reliability from previous assignments, where ensuring performance under load was identified as critical for maintaining user satisfaction.
- **User Story:** *As a user*, I want the system to return results within 2 seconds for common tasks, so I can complete my work without unnecessary delays.
  - *Reference:* Drawing on performance measures discussed in the scope of the project, this reinforces the need for a swift, responsive system.

### 4. Operational and Environmental Requirements

- **User Story:** *As a user*, I want to be able to access the product at any time, 24 hours a day, 7 days a week, so that I am not limited by time constraints when performing critical work (RFP Point 7).
  - *Reference:* This ties to the product's availability requirement highlighted in earlier assignments, where 24/7 access was considered essential for maintaining user engagement.

- **User Story:** *As a system administrator*, I want the product to be compatible with the latest versions of all major browsers (Chrome, Firefox, Edge, etc.), so that users can access the product from any device without compatibility issues (RFP Point 10).
  - *Reference:* Reflecting on the technical specifications discussed previously, cross-browser compatibility was emphasized as a critical feature to ensure a broad user base.

## 5. Maintainability and Support Requirements

- **User Story:** *As an IT manager*, I want the product to require only a maximum of 2 hours of maintenance every 3 months, ensuring that users experience minimal disruption during system updates (RFP Point 9).
  - *Reference:* This was directly derived from the need for regular maintenance discussed in Assignment 1, ensuring a balance between product stability and ongoing support.

## 6. Security Requirements

- **User Story:** *As a user*, I want the product to use advanced encryption for all personal data so that I feel confident my sensitive information is fully protected from breaches (RFP Point 3).
  - *Reference:* Security concerns were a top priority in the earlier assignments, with data protection mechanisms being one of the product's core non-functional requirements.
- **User Story:** *As a user*, I want the system to include two-factor authentication to safeguard my account, ensuring that even if my password is compromised, my data remains secure (RFP Point 11).
  - *Reference:* This user story is built upon the security best practices discussed in earlier requirements gathering sessions, highlighting the importance of multiple layers of protection.

## 7. Cultural Requirements

- **User Story:** *As a bilingual user*, I want to seamlessly switch between English and French in the product's interface, ensuring that I can use the product in my preferred language for convenience and accuracy (RFP Point 6).
  - *Reference:* This requirement connects to the need for a localized experience, which was stressed as a part of the product's global usability goals.

## 8. Legal Requirements

- **User Story:** *As a compliance officer*, I want the product to comply with international data protection laws, such as GDPR, so that our company avoids legal risks and ensures users' rights are respected (RFP Point 12).
  - *Reference:* Legal compliance was noted as a major concern in earlier discussions, particularly in relation to privacy laws and the safe handling of personal data.

# Functional Requirements User Stories

1. **User Story:** *As a user*, I want to securely log into my account using multi-factor authentication, ensuring that only authorized individuals can access sensitive system functions (RFP Point 11 - Authorization).
  - *Reference:* This is based on the importance of strong access control mechanisms discussed in previous assignments.
2. **User Story:** *As a user*, I want to interact with the system in either English or French, depending on my preference, to ensure accessibility and usability (RFP Point 6).
  - *Reference:* This requirement for multilingual support reflects the importance of inclusivity and cultural adaptation, which was emphasized earlier.
3. **User Story:** *As a user*, I want the product interface to be intuitive, with easy navigation and minimal steps to complete tasks, so that I can efficiently perform my work without requiring advanced training (RFP Point 2).
  - *Reference:* The focus on usability and reducing friction for users was a central theme in the requirements outlined in Assignment 1.
4. **User Story:** *As an auditor*, I want to be able to view and analyze detailed audit logs, so that I can verify financial transactions and ensure compliance with corporate policies (RFP Point 12).
  - *Reference:* This draws from the earlier discussion about ensuring traceability and transparency within the product.
5. **User Story:** *As a user*, I want the system to maintain performance even during periods of high traffic, ensuring that I can work uninterrupted without system failures (RFP Point 8).
  - *Reference:* The need for fault tolerance was highlighted in the previous assignments to ensure system stability during heavy usage.
6. **User Story:** *As a user*, I want the system to be accessible at any time of day or night, without interruptions, to support my ongoing work (RFP Point 7).
  - *Reference:* This aligns with the product's 24/7 availability requirement, which was discussed earlier to ensure a seamless user experience.
7. **User Story:** *As a system administrator*, I want to assign user roles and manage permissions, so that I can control who has access to sensitive data and actions within the system (RFP Point 11).
  - *Reference:* This is in line with the need for granular user management and security discussed in prior assignments.
8. **User Story:** *As a user*, I want a simple and secure process to recover my password if I forget it, ensuring that I can quickly regain access to my account without having to contact support (RFP Point 11).
  - *Reference:* Password recovery was identified as a critical part of user management and system security in earlier assignments.
9. **User Story:** *As a user*, I want to be able to access the system from any device or browser, so that I can use the product regardless of my preferred technology (RFP Point 10).
  - *Reference:* Cross-browser compatibility was emphasized as part of ensuring the product is accessible to all users, regardless of their platform.

10. **User Story:** *As an IT administrator*, I want system updates and maintenance to require minimal downtime (no more than 2 hours every 3 months), so that users experience no significant disruptions during routine maintenance (RFP Point 9).
  - *Reference:* This user story aligns with the maintainability goals discussed previously, ensuring continuity of operations.
11. **User Story:** *As a user*, I want my personal and financial data to be securely encrypted so that I can trust the system to protect my sensitive information (RFP Point 3).
  - *Reference:* Data security was a top concern in earlier discussions and is carried forward in this user story, focusing on encryption.
12. **User Story:** *As a system administrator*, I want to monitor system performance in real-time, so that I can address any issues promptly and ensure the system operates efficiently under heavy usage (RFP Point 8).
  - *Reference:* This emphasizes the need for proactive monitoring, a key requirement identified earlier for system reliability.